KUBERNETES NETWORKING FUNDAMENTALS

NETWORKING FLOW

Cluster ------> worker node ------> pods ------> container --------> application

Application runs on a specific port inside the Container
↓
Container Port is exposed through the Pod
↓
Pod gets its own IP Address inside the Cluster
↓
Service exposes the Pod IP and Port
↓
NodePort/LoadBalancer exposes the Service outside the Cluster
↓
User accesses the Application using:
NodeIP:NodePort
or
LoadBalancer DNS/IP

SERVICES in KUBERNETES

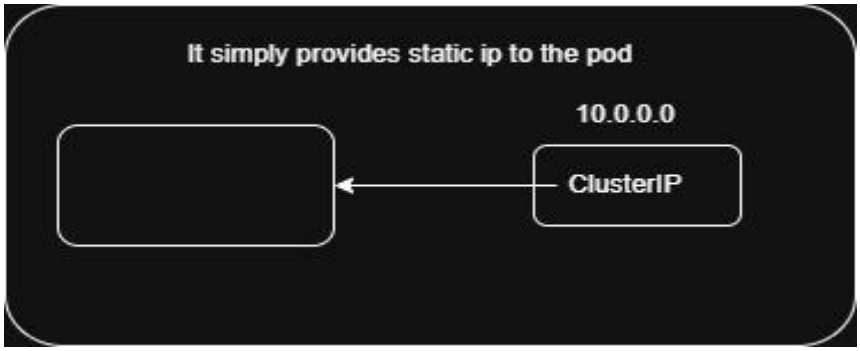
A Kubernetes Service is used to expose Pods outside of the cluster and provide stable networking.

Types Of Services In Kubernetes

1. ClusterIP
2. NodePort
3. LoadBalancer

1. ClusterIP (Default)

- Exposes application only inside the cluster
- Cannot access from outside
- Provides a stable/static virtual IP address for communication
- Used for internal communication between Pods and Services
- And it’s a Default service type in Kubernetes

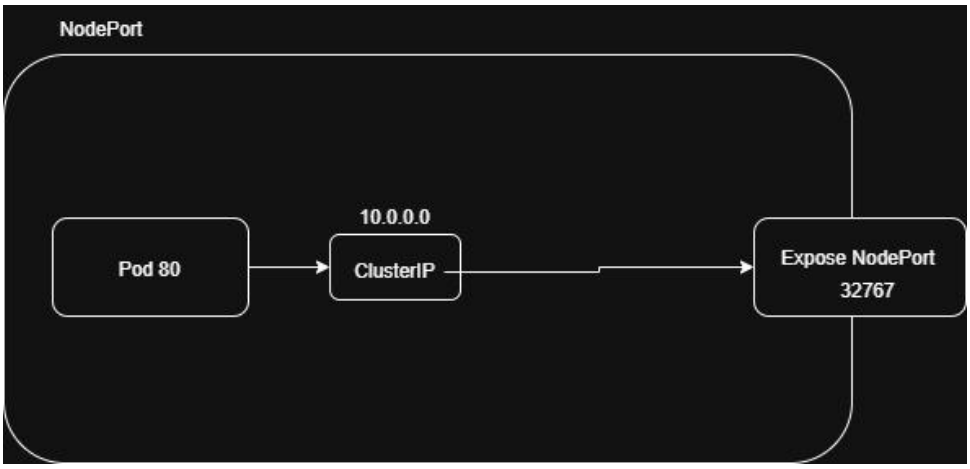


1. kubectl run cipod --image=nginx --port=80
2. kubectl expose pod cipod --name=cipod-svc --type=ClusterIP --target-port=80 --port=80
3. kubectl get pods -o wide
   curl <ip_pod>
4. kubectl get svc -o wide
5. kubectl delete pod cipod

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2. NodePort

- Exposes application outside the Kubernetes Cluster
- Application can be accessed using NodeIP:NodePort
- Its range started from 30000–32767



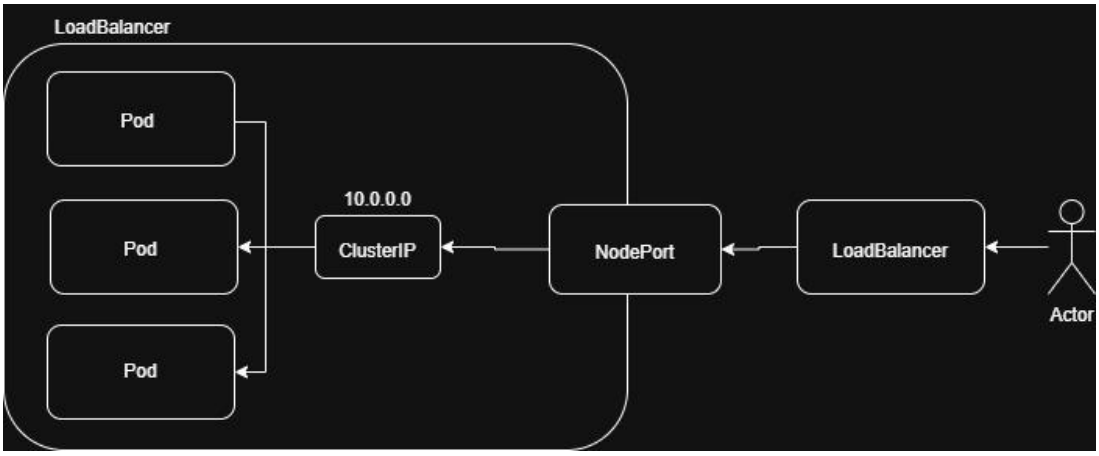
1. kubectl run nppod –image=nginx –port=80
2. kubectl expose pod nppod --name=nppod-np --type=NodePort --target-port=80 --port=80
3. kubectl get svc -o wide
4. kubectl get nodes -o wide
   curl <node_ip:expose_port>

5. kubectl delete pod nppod

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3. LoadBalancer

- Exposes application using an external cloud Load Balancer (DNS)
- Automatically distributes traffic across Pods



1. kubectl run lbpod –image=nginx –port=80
2. kubectl expose pod lbpod --name=lbpod-lb --type=LoadBalancer --port=80 --target-port=80
3. kubectl get nodes -o wide
4. kubectl get svc -o wide
   <dns_name> ----> hit on web
5. kubectl delete pod lbpod

COMMANDS

- kubectl get svc -o wide
- kubectl describe pod <pod_name>

DIFFERNCE BETWEEN INTRA-POD & INTER-POD COMMUNICATION

INTRA-POD

- Communication happens between containers inside the same Pod.
- All containers share the same IP address and network namespace.
- Containers communicate using localhost.
- Mainly used for sidecar and helper container communication.

INTER-POD

- Communication happens between different Pods in the cluster.
- Each Pod has its own unique IP address.
- Pods communicate using Pod IP or Kubernetes Services.
- Mainly used for communication between microservices applications.
